

Literature Answer To Rosendal's $\text{Isom}(\mathbb{Q}\mathbb{U})$ Reflexive Fixed-Point Question

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Status

Status: literature already answered. This packet records an exact later literature answer to Problem 66 of Christian Rosendal's *Large scale geometry of metrisable groups*, arXiv:1403.3106.

Source Question

In arXiv:1403.3106, Problem 66 on page 39 asks whether $\text{Isom}(\mathbb{Q}\mathbb{U})$, the isometry group of the rational Urysohn metric space, has a continuous affine isometric action on a reflexive Banach space without a fixed point.

Answering Theorem

The later paper Christian Rosendal, *Equivariant geometry of Banach spaces and topological groups*, arXiv:1610.01070, answers this question negatively. Corollary 12 on page 5 states the fixed-point conclusion for $\text{Isom}(\mathbb{Q}\mathbb{U})$. The detailed proof is Theorem 81 on pages 47–48: if E is reflexive, or $E = L^1([0, 1])$, and

$$G = \text{Isom}(\mathbb{Z}\mathbb{U}) \quad \text{or} \quad G = \text{Isom}(\mathbb{Q}\mathbb{U}),$$

then every quasi-cocycle $b : G \rightarrow E$ is bounded. In particular, every affine isometric action $G \curvearrowright E$ has a fixed point.

Taking $G = \text{Isom}(\mathbb{Q}\mathbb{U})$ and E reflexive gives exactly the answer to Problem 66: no such fixed-point-free continuous affine isometric action exists.

Scope

This packet answers only Problem 66 of arXiv:1403.3106. The broader Problem 67 from that source, concerning all non-Archimedean Polish groups, is not settled by this identification. Also, arXiv:1610.01070 asks a stronger Question 82 in which the action is affine but not assumed isometric; that stronger non-isometric variant remains outside this packet.

Provenance

The answer is not original to this run. The run contribution is only the literature identification: a separate later paper by Rosendal contains the exact fixed-point theorem answering the earlier extracted question.

References

- Christian Rosendal, *Large scale geometry of metrisable groups*, arXiv:1403.3106.
- Christian Rosendal, *Equivariant geometry of Banach spaces and topological groups*, arXiv:1610.01070.